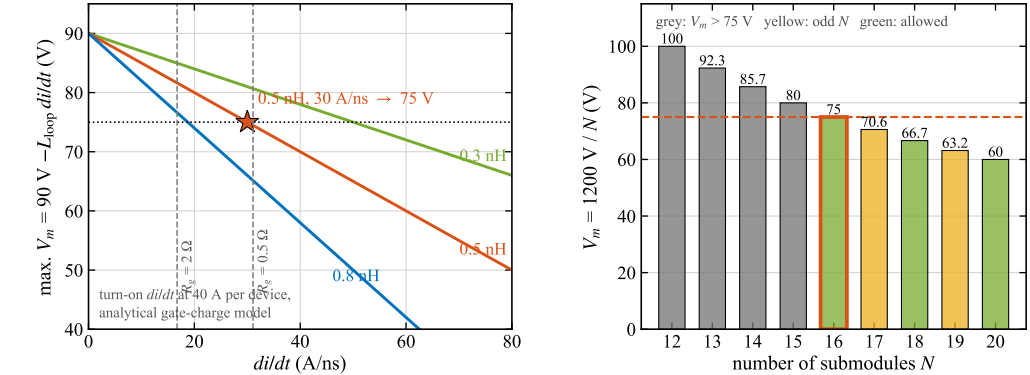
max. $V_m = 90\ \text{V} - L_{\text{loop}}\, di/dt$ (V)
di/dt (A/ns)
0.5 nH, 30 A/ns → 75 V
0.3 nH
0.5 nH
0.8 nH
$R_g = 2\ \Omega$
$R_g = 0.5\ \Omega$
turn-on di/dt at 40 A per device,
analytical gate-charge model
grey: $V_m > 75$ V yellow: odd N green: allowed
$V_m = 1200\ \text{V} / N$ (V)
number of submodules N
100
92.3
85.7
80
75
70.6
66.7
63.2
60

(a) (b)